

Arnav Tiwari

Full Stack Software Engineer

Next.js • React • Node.js • PostgreSQL • Django • AI-Powered Applications

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Professional Summary

Full-Stack Software Engineer with hands-on experience developing AI-powered applications, industrial software, and security platforms. Delivered measurable business impact through production-ready solutions, including reducing hazardous incidents by 54% at Hitachi India, accelerating document workflows by 80%, and improving application performance by up to 50% at Payatu Security. Experienced in building scalable React/Next.js frontends, Node.js and Django backends, REST APIs, SQL/NoSQL databases, and integrating LLM-powered AI workflows.

Technical Skills

Languages: C, C++, Java, Python, JavaScript, TypeScript

Frontend: React.js, Next.js, Redux, HTML5, CSS3, Tailwind CSS, SCSS

Backend & APIs: Node.js, Express.js, Django, FastAPI, REST, GraphQL, JWT, OAuth 2.0, WebSockets

Databases: MongoDB, PostgreSQL, MySQL, SQLAlchemy, Drizzle ORM

DevOps & Cloud: Docker, Kubernetes, CI/CD, GitHub Actions, Git, Linux, Bash, Vercel (Cloud Deployment)

CS Fundamentals: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Professional Experience

Software Developer Intern

July 2025 – June 2026

Hitachi India Pvt Ltd. | Nagpur, Maharashtra

- Engineered SLD visualization modules for steel plant electrical layouts, enabling real-time monitoring of motors and critical equipment — directly contributing to a **33% reduction in motor failures** and **54% reduction in hazardous incidents**.
- Built React + Django interfaces for 5 HICADS systems (DPS, RCTMS, PTMS, TTMS, ICQIS), digitalizing manual electrical and steel equipment analysis workflows, reducing operator time consumption by **57%** and improving work efficiency by **43%**.
- Automated CRS document generation pipeline using n8n, reducing turnaround from **2 weeks to 2 days (80% faster)** and enabling delivery of **7–10 project documents weekly** that previously took a month.

Full Stack Developer Intern – MERN Stack

May 2025 – July 2025

Payatu Security Consulting Pvt. Ltd. | Pune, Maharashtra

[Certificate](#)

- Built responsive Next.js + SCSS interfaces reducing page load time from **3s to 1.5s (50% faster)**, improving UI maintainability and user experience.
- Developed 15 REST API endpoints and debugged and optimized 8 using indexing, query optimization, and caching on a Node.js + Express.js backend, improving response time by **40%** in collaboration with cross-functional teams.
- Built a real-time logger handling **1,000+ log events/minute** for a 4-role RBAC (Admin, System Engineer, Technician, Employee) WiFi admin dashboard managing **1,000+ users**.

Projects

Autopilot – AI Powered Project Planning Platform

[Link](#) | [GitHub](#)

Next.js, TypeScript, React.js, Node.js, Express.js, PostgreSQL, MongoDB, Google Gemini, Groq SDK, OAuth 2.0, Vercel

- Architected a full-stack MERN-style application with a decoupled multi-agent AI system using 4 specialized parallel agents (Gemini, Groq SDK), reducing project plan generation from **10–15 minutes to under 3 minutes (83% faster)** vs manual workflows.
- Designed a RESTful API layer with **18 Node.js/Express.js endpoints** orchestrating agent coordination and returning structured execution plans with confidence scores, cost estimations, and task breakdowns.
- Delivered OAuth 2.0 (Google + GitHub) auth, PostgreSQL integration, multi-format export (PDF, DOCX, JSON), i18n, and WCAG-compliant dark-mode UI deployed on Vercel cloud.

SkillBridge – AI Adaptive Onboarding Engine

[Link](#) | [GitHub](#)

React, FastAPI, Python, FAISS, spaCy, Sentence-Transformers, NetworkX, Docker, SQLAlchemy

- Built an AI-driven skill gap analyzer processing resumes against job descriptions using a **3-layer matching pipeline** (exact match, skill family, cosine similarity via FAISS) trained on **2,484 resumes and 2,277 JDs** achieving **skill extraction F1 > 0.82** and **gap detection recall > 85%**.
- Designed an original **Weighted Graph Traversal (WGT) algorithm** with Kahn’s topological sort for prerequisite-aware learning pathway generation across a **71-skill taxonomy and 58-module** curated course catalog.
- Delivered full-stack solution with React DAG visualization (react-flow), real-time 6-stage processing pipeline, LLM reasoning traces, and Docker deployment – generating personalized roadmaps **in under 100 seconds**.

ContentForge AI – AI Content Generation SaaS

[Link](#) | [GitHub](#)

Next.js, React.js, Node.js, TypeScript, Clerk, Drizzle ORM, PostgreSQL, Tailwind CSS, Razorpay, Google Gemini API

- Engineered a scalable LLM-powered SaaS platform with **18 dynamic content templates** spanning blog posts, ad copy, social media, and SEO content, reducing content creation time by **70%** vs manual workflows.
- Built Node.js + PostgreSQL backend with Drizzle ORM featuring schema-level data integrity, query optimization, and full CRUD support for AI generation history and user data.
- Implemented end-to-end authentication via Clerk with usage-based credit tracking, and Razorpay subscription billing with webhook handling for automated plan management.

Achievements

- Rated **5★ (rating 2029)** on CodeChef, ranking among top competitive programmers globally | Solved **400+ DSA** problems on LeetCode.
- 1st Runner Up – ArtPark Codeforge Hackathon** (Unstop) – competed against teams across India in a software development challenge.
- Finalist (Top 2.6% of 31,000+ Teams)**, OpenEnv Hackathon × Scaler School of Technology – Advanced through multiple rounds to the 36-hour onsite finals in Bangalore (Apr 2026).

Education

Indian Institute of Information Technology Nagpur (IIIT Nagpur)

November 2022 – July 2026

BTech in Computer Science Engineering | **CGPA: 7.58/10**